

1. (1) 破色子 $C_6' + C_6^2 = 6 + 15 = 21$

(2) 费米子 $C_6^2 = 15$

(3) 可区分 $6 \times 6 = 36$

2. $S_{\text{bin}}(p) = -p \log p - (1-p) \log (1-p)$

则 $\frac{dS_{\text{bin}}(p)}{dp} = -\log p - \frac{1}{\ln 2} + \log(1-p) + \frac{1}{\ln 2}$

$= -\log p + \log(1-p) = \log \frac{1-p}{p} = 0$

故 $p = \frac{1}{2}$, $S_{\text{bin}}(p)$ 在 $p = \frac{1}{2}$ 时取最大值得证。

3. (1) $|\lambda I - A| = 0$

$p = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

$(\lambda - 1) \cdot \lambda = 0$, 则 $\lambda_1 = 1, \lambda_2 = 0$.

$S(p) = -\sum_x \lambda_x \log \lambda_x = -\log 1 - 0 \log 0 = 0$

(2) $p = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

$(\lambda - \frac{1}{2})^2 - (\frac{1}{2})^2 = 0$, 则 $\lambda_1 = 1, \lambda_2 = 0$

$S(p) = -\sum_x \lambda_x \log \lambda_x = -\log 1 - 0 \log 0 = 0$

$$(3) \quad p = \frac{1}{3} \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

$$(\lambda - \frac{2}{3})(\lambda - \frac{1}{3}) - (\frac{1}{3})^2 = 0$$

$$R_1) \quad \lambda_1 = \frac{1}{2} + \frac{\sqrt{5}}{6}, \quad \lambda_2 = \frac{1}{2} - \frac{\sqrt{5}}{6}$$

$$S(p) = -\sum_x \lambda_x \log \lambda_x$$

$$= -(\frac{1}{2} + \frac{\sqrt{5}}{6}) \log(\frac{1}{2} + \frac{\sqrt{5}}{6}) - (\frac{1}{2} - \frac{\sqrt{5}}{6}) \log(\frac{1}{2} - \frac{\sqrt{5}}{6})$$

$$\approx 0.55$$

选择:

1. c

2. d

3. d

4. abcd

5. d

6. c

7. f

8. e

9. h

10. efgh